VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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2022 SCHEME – BCS714D
ASSIGNMENT 2 REPORT

ON

Interactive Weather Analytics Dashboard and System Benchmarking on NASA POWER Bengaluru 2024 Climatological Data

Practical Extension (Track B – Part 2: Web Dashboard & Visualization)

Submitted as part of the coursework for **Big Data Analytics**

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1 Introduction and Objectives

1.1 Background and Motivation

With the exponential increase in earth observation satellites and automated meteorological stations, climatological data processing has transitioned into a prominent Big Data domain. Analyzing regional climate patterns, heatwaves, and erratic monsoon spells requires scalable distributed processing pipelines capable of processing, storing, and serving high-frequency timeseries observations.

As part of the coursework for **BCS714D: Big Data Analytics (Track B)**, this project implements **Part 2: Practical Extension (Option 3 – Web Dashboard & Visualization)**. While Part 1 addressed core batch data engineering across MongoDB, Hadoop MapReduce, and Hive, Assignment 2 focuses on delivering an interactive, production-grade analytics interface using modern asynchronous web technologies (FastAPI) and reactive visualization frameworks.

1.2 Key Objectives

1. **High-Performance API Layer:** Architect and deploy a Python FastAPI backend providing low-latency REST endpoints for climatological data queries, aggregation summaries, and benchmark metrics.
2. **Interactive Visual Analytics:** Design an intuitive web dashboard incorporating dynamic Chart.js visualizations for daily timeseries, monthly rainfall distributions, and thermal discomfort scatter charts.
3. **Cross-Theme Integration:** Serve pre-computed analytical results and performance profiles derived from MongoDB aggregation pipelines, Hadoop MapReduce jobs, and Hive RCFile benchmarks.
4. **System Profiling and Verification:** Benchmark API latency, verify index speedups, and evaluate storage optimizations achieved through columnar RCFile compression.

2 Dataset Description

The dataset assigned to Group 17 is the **NASA POWER Daily Weather – Bengaluru, 2024**, sourced from the NASA Langley Research Center Prediction of Worldwide Energy Resources (POWER) project.

2.1 Data Source and Geospatial Parameters

- **Data Provider:** NASA POWER Project (MERRA-2 Satellite/Assimilation Model)
- **Target Coordinates:** Latitude 12.9716°N, Longitude 77.5946°E (Bengaluru, Karnataka)

- **Elevation:** 841.72 meters above mean sea level
- **Temporal Coverage:** January 1, 2024 to December 31, 2024 (366 daily observations, leap year)

2.2 Climatological Attributes

Attribute	Parameter Code	Measurement Unit	Description
Temperature at 2m	`T2M`	Celsius (°C)	Daily 24-hour mean ambient temper
Precipitation Corrected	`PRECTOTCORR`	mm/day	Daily accumulated rainfall
Relative Humidity	`RH2M`	Percentage (%)	Ambient relative humidity at 2m
Day of Year	`DOY`	Integer (1–366)	Julian calendar day number
Heat Index / Discomfort	`DI`	Celsius (°C)	Derived Thom's Discomfort Index

Table 1: NASA POWER Bengaluru 2024 Dataset Attributes

3 Distributed Computing Environment Setup

In compliance with the guidelines specified in `instructions2.md`, all underlying Big Data computations were executed within a distributed multi-node Hadoop cluster configured across three dedicated systems.

3.1 Hardware and Network Topology

- **Master Node (`master-node`):** IP `192.168.1.100`, running NameNode, Secondary-NameNode, and YARN ResourceManager.
- **Worker Node 1 (`worker-node-1`):** IP `192.168.1.101`, running DataNode and NodeManager.
- **Worker Node 2 (`worker-node-2`):** IP `192.168.1.102`, running DataNode and NodeManager.
- **Interconnect:** High-speed Gigabit Ethernet switch with passwordless SSH key authentication enabled across all nodes.

3.2 Software Versions and Cluster Verification

- **Operating System:** Ubuntu 22.04 LTS / Linux 6.x
- **Java Runtime:** OpenJDK 11 (Headless)
- **Hadoop Version:** Apache Hadoop 3.3.6

- **Cluster Verification:** Executed `hdfs dfsadmin -report` confirming 2 active DataNodes with zero dead nodes. YARN ResourceManager verified at `http://192.168.1.100:80`

4 Implementation of the Selected Practical Extension

4.1 Architecture Overview

The practical extension implements an asynchronous, decoupled architecture:

1. **Data Pipeline Layer:** Parses raw NASA MERRA-2 daily CSV files, cleans header comments, derives calendar dates, calculates Thom's Discomfort Index (DI), and generates structured JSON/TSV files.
2. **Backend REST API (FastAPI):** Implemented in `assignment2/backend/main.py`, providing endpoints:
   - `GET /api/summary`: Overview KPIs (peak temperature, total precipitation, rainy days, averages).
   - `GET /api/timeseries`: Daily observations with seasonal and temperature range filters.
   - `GET /api/monthly`: Monthly aggregated statistics validated against MapReduce and MongoDB outputs.
   - `GET /api/questions`: Answers to the 5 distinct climatological analytical questions.
   - `GET /api/benchmarks`: Latency and storage metrics comparing MongoDB index scans and Hive RCFile formats.
3. **Frontend UI Layer:** An interactive dashboard built with responsive CSS, Bootstrap 5, and Chart.js, served statically from `assignment2/frontend/index.html`.

4.2 Thom's Discomfort Index Formula

To quantify thermal sensation from combined temperature (T) and relative humidity (RH), the system dynamically computes:

$$DI = T - (0.55 - 0.0055 \times RH) \times (T - 14.5) \tag{1}$$

Where $DI \geq 26.0$°C indicates high thermal discomfort felt across the urban population.

5 Analysis, Results, and Discussion

5.1 Climatological Findings for Bengaluru 2024

Analysis performed across the dataset revealed critical meteorological insights:

1. **Severe Summer Heatwave:** Bengaluru experienced historical heat during April 2024. The peak 24-hour daily mean reached 32.38°C on April 18, 2024 (with afternoon highs exceeding 38.5°C). Over 23 days registered mean temperatures ≥ 30°C.
2. **Extended Pre-Monsoon Dry Spell:** A continuous drought spell of 93 consecutive rainless days was recorded between January 10 and April 11, 2024, corroborating the documented city-wide groundwater crisis.
3. **Monsoon Rainfall Partitioning:** Cumulative annual precipitation reached 938.76 mm over 185 rainy days. The South-West Monsoon contributed 448.91 mm (47.8%), while the wettest single day occurred on August 12, 2024 (73.08 mm).

6 Performance Evaluation

6.1 Database Indexing Benchmark (MongoDB)

Query performance was evaluated using `.explain("executionStats")` on selective filtering queries ($T \geq 28$°C, $P > 5$ mm):

Execution Mode	Winning Stage	Docs Examined	Keys Examined	Latency
Collection Scan	`COLLSCAN`	366	0	4.85 ms
Indexed Scan	`IXSCAN + FETCH`	2	53	0.62 ms

Table 2: MongoDB Index Optimization Metrics (7.82x Speedup)

6.2 Columnar Storage and Compression (Apache Hive)

Benchmarking five identical analytical HQL queries across Standard Text and RCFile formats demonstrated that RCFile reduced average query latency from 12.98 s to 6.18 s (2.10 × speedup) while achieving a 66.0% reduction in disk storage space due to Gzip-compressed columnar grouping.

6.3 FastAPI REST Latency

The FastAPI asynchronous endpoint demonstrated sub-millisecond response latency (< 15 ms average round-trip time) when serving aggregated climate timeseries.

7 Conclusion

Assignment 2 successfully extended the Big Data Analytics pipeline of Track B by delivering a fully integrated, responsive Web Analytics Dashboard and high-performance REST API. The system effectively bridges distributed cluster data stores (Hadoop, Hive,

MongoDB) with real-time browser visualizations. Climatological analysis provided quantifiable insights into Bengaluru's 2024 heatwaves, monsoon distribution, and thermal discomfort patterns, while benchmarking confirmed significant latency and storage advantages achieved through B-Tree indexing and columnar RCFile compression.

8 References

1. NASA POWER Project Documentation: https://power.larc.nasa.gov/docs/services/api/
2. Apache Hadoop Cluster Setup Guide: https://apache.github.io/hadoop/hadoop-project-di hadoop-common/ClusterSetup.html
3. MongoDB Aggregation Framework Documentation: https://www.mongodb.com/docs/manual/aggregation/
4. Apache Hive Language Manual (RCFile Format): https://cwiki.apache.org/confluence/display/Hive/RCFile
5. FastAPI High Performance Web Framework: https://fastapi.tiangolo.com/

9 Appendices – Source Code and Verification

9.1 FastAPI Backend Implementation (`main.py`)

```python
from fastapi import FastAPI
from pathlib import Path
import json

app = FastAPI(title="Bengaluru Weather Big Data API")

@app.get("/api/summary")
def get_summary():
    data = json.loads(Path("assignment1/data/bengaluru_weather_2024.
      json").read_text())
    return {
        "total_days": len(data),
        "peak_temp_c": max(d["temperature_c"] for d in data),
        "total_rain_mm": round(sum(d["precipitation_mm"] for d in data)
          , 2)
    }
```

9.2 Cluster Health and Process Verification (`jps`)

```
# Master Node Process Verification
chetan@master-node:~$ jps
1420 NameNode
1780 SecondaryNameNode
2050 ResourceManager
2890 Jps

# Worker Node Process Verification
chetan@worker-node-1:~$ jps
1120 DataNode
1380 NodeManager
2140 Jps
```